

[2]

✓ 16s

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) Advertising.csv**Advertising.csv**(text/csv) - 5166 bytes, last modified: 5/4/2023 - 100% done

Saving Advertising.csv to Advertising.csv



[3]

✓ 0s

```
import pandas as pd

df = pd.read_csv("Advertising.csv")
df.head()
```



	Unnamed: 0	TV	Radio	Newspaper	Sales
0	1	230.1	37.8	69.2	22.1
1	2	44.5	39.3	45.1	10.4
2	3	17.2	45.9	69.3	9.3
3	4	151.5	41.3	58.5	18.5
4	5	180.8	10.8	58.4	12.9



Next steps:

[Generate code with df](#)[New interactive sheet](#)

[4]

✓ 0s

```
print(df.info())
print(df.describe())
```



<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	Unnamed: 0	200 non-null	int64
1	TV	200 non-null	float64
2	Radio	200 non-null	float64
3	Newspaper	200 non-null	float64
4	Sales	200 non-null	float64

dtypes: float64(4), int64(1)

memory usage: 7.9 KB

None

	Unnamed: 0	TV	Radio	Newspaper	Sales
count	200.000000	200.000000	200.000000	200.000000	200.000000
mean	100.500000	147.042500	23.264000	30.554000	14.022500
std	57.879185	85.854236	14.846809	21.778621	5.217457
min	1.000000	0.700000	0.000000	0.300000	1.600000
25%	50.750000	74.375000	9.975000	12.750000	10.375000
50%	100.500000	149.750000	22.900000	25.750000	12.900000
75%	150.250000	218.825000	36.525000	45.100000	17.400000
max	200.000000	296.400000	49.600000	114.000000	27.000000



[5]

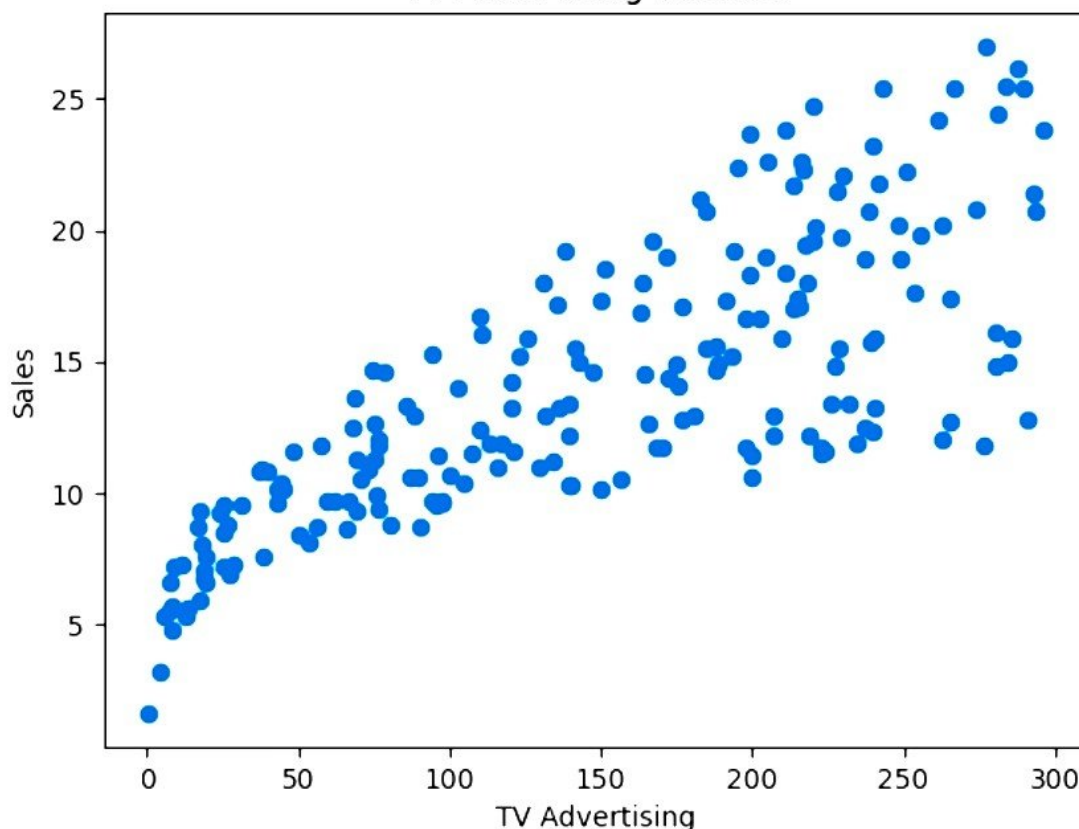
✓ 0s

```
import matplotlib.pyplot as plt

plt.scatter(df['TV'], df['Sales'])
plt.xlabel("TV Advertising")
plt.ylabel("Sales")
plt.title("TV Advertising vs Sales")
plt.show()
```



TV Advertising vs Sales



[7]

✓ 1s

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X = df[['TV', 'Radio', 'Newspaper']]
y = df['Sales']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
```



LinearRegression

LinearRegression()



[8]
✓ 0s

```
from sklearn.metrics import r2_score  
  
y_pred = model.predict(X_test)  
  
print("R2 Score:", r2_score(y_test, y_pred))
```



R2 Score: 0.899438024100912



[9]
✓ 0s

```
sample = [[200, 50, 30]]  
  
prediction = model.predict(sample)  
  
print("Predicted Sales:", prediction[0])
```



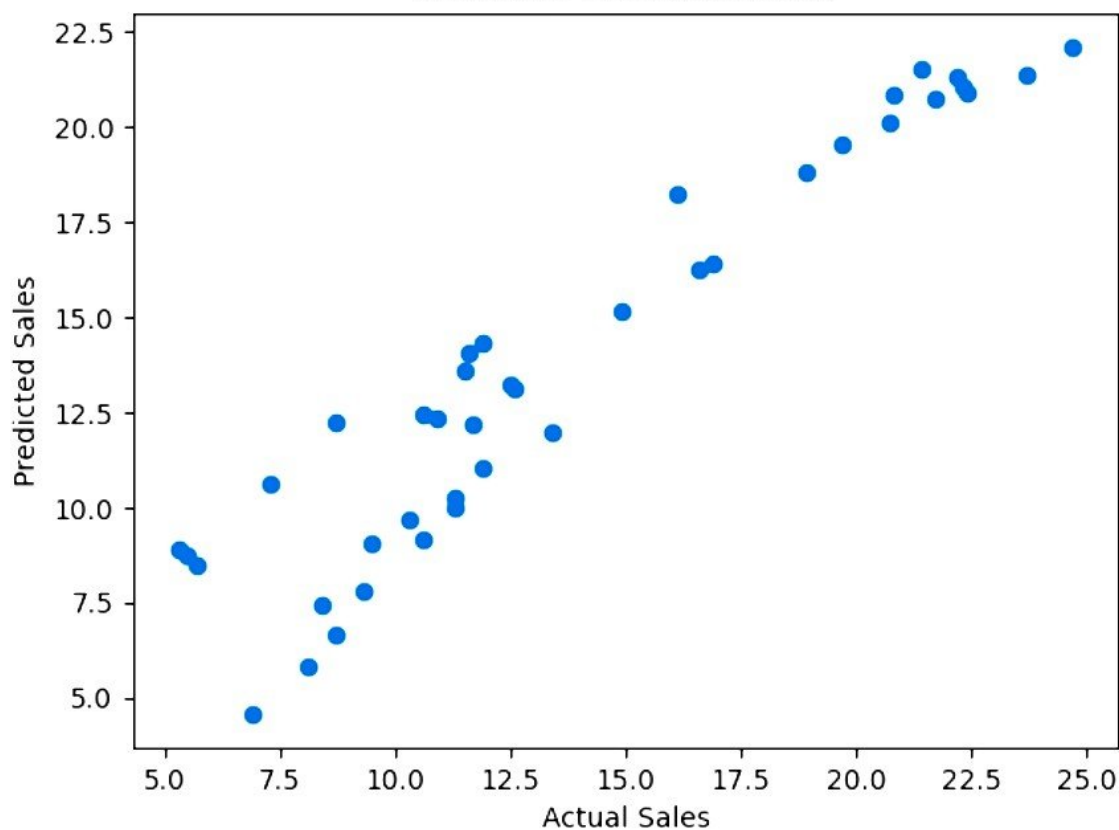
Predicted Sales: 21.467556973825737
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739
warnings.warn(

[10]
✓ 0s

```
import matplotlib.pyplot as plt  
  
plt.scatter(y_test, y_pred)  
plt.xlabel("Actual Sales")  
plt.ylabel("Predicted Sales")  
plt.title("Actual vs Predicted Sales")  
plt.show()
```



Actual vs Predicted Sales





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Close



****INSIGHT****

1. TV advertising has a strong impact on sales.
2. Higher advertising spend generally leads to increased sales.
3. The Linear Regression model successfully predicted sales values.
4. The model showed good performance based on the R^2 score.
5. Data-driven predictions can help businesses improve marketing decisions.

INSIGHT

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